Anal Roy Chowdhury

Ph.D. Student · School of Mathematical & Computational Sciences · IACS, Kolkata

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Homepage | Google Scholar | GitHub | LinkedIn | Twitter/X

Research Interests

Multimodal & Multi-view Learning • Multiple Kernel Methods • Sign Language Processing • Computer Vision • Natural Language Processing • Generative Neural Networks • Graph Neural Networks • Unsupervised & Self-supervised Learning • Deep Clustering

Education

Ph.D. in Mathematical & Computational Sciences — Aug 2023 – Present

Indian Association for the Cultivation of Science (IACS), Kolkata

Supervisor: Dr. Debarshi Kumar Sanyal · Focus: Multimodal Learning, Sign Language Processing

M.Sc. in Electronic Science — 2016 – 2018

Acharya Prafulla Chandra College, West Bengal State University, Kolkata

Thesis: Analytical Computation of Electrical Properties of ID-DG & TG-DG MOSFETs following Ortiz-Conde Model · Supervisor: Dr. Arpan Deyasi

B.Sc. in Physics (Hons.) — 2013 – 2016

Seth Anandram Jaipuria College, University of Calcutta, Kolkata

Research Experience

Ph.D. Research Scholar — Aug 2023 – Present

SMCS, Indian Association for the Cultivation of Science (IACS), Kolkata

- Conducting research on multimodal learning with a special focus on Sign Language Processing (SLP), especially translation of Indian Sign Language (ISL), under the supervision of Dr. Debarshi Kumar Sanyal.
- Developing motion-aware and vision–language models for real-time ISL processing, targeting the deaf and hard-of-hearing community in India.
- Published at **ACL 2026** (Findings) and **WSLP @ AACL-IJCNLP 2025**.

Project Linked Personnel — Jul 2021 – Mar 2023

Electronics & Communication Sciences Unit (ECSU), Indian Statistical Institute (ISI), Kolkata

- Worked on multiple kernel approaches for efficient multi-view clustering under Prof. Swagatam Das.
- Conducted a comprehensive survey of deep multi-view clustering methods, published in *Information Fusion* (Elsevier, 2025).
- Research areas: Multiple Kernel Metric Learning, Deep Clustering, Multi-view Clustering.

Research Project Assistant — Feb – Dec 2019

Sangutech Enterprises Pvt. Ltd., Kolkata (DBT-BIRAC funded, Govt. of India)

- Contributed to an agonist-free point-of-care platelet function testing project, incubated at CRNN, University of Calcutta, under Dr. Suryyani Deb.
- Designed deep learning pipelines for biomedical image analysis in collaboration with Dr. Ashis Kumar Dhara (NIT Durgapur).
- Co-invented a UV-light-based platelet aggregation detection system (Multi-scale FCN + deep neural network) → **Indian Patent IN580844 B1**.

Publications

[1] **Anal Roy Chowdhury** and Debarshi Kumar Sanyal. "Can Small Vision–Language Models Perform Sign Language Translation?" In *Findings of the Association for Computational Linguistics: ACL 2026*, San Diego, California. ACL, **2026**.

[2] **Anal Roy Chowdhury**, Avisek Gupta, and Swagatam Das. "Deep multi-view clustering: A comprehensive survey of the contemporary techniques." *Information Fusion*, Elsevier, p. 103012, **2025**. [doi]

[3] **Anal Roy Chowdhury** and Debarshi Kumar Sanyal. "Enhancing Indian Sign Language Translation via Motion-Aware Modeling." In *Proceedings of the Workshop on Sign Language Processing (WSLP)*, pp. 39–46, IIT Bombay, Mumbai (co-located with IJCNLP–AACL 2025). ACL, **2025**. [doi]

[4] K. Roy, **A.R. Chowdhury**, and A. Deyasi. "Computation of Gate-Induced-Drain-Leakage Current Due to Band-to-Band Tunneling for Ultrathin MOSFET." In *Photonics and Communication. Lecture Notes in Networks and Systems*, Springer, vol. 79, pp. 3–10, **2020**. [doi]

[5] A. Deyasi, A. Sarkar, K. Roy, and **A.R. Chowdhury**. "Effect of high-K dielectric on differential conductance and transconductance of ID-DG MOSFET following Ortiz-Conde model." *Microsystem Technologies*, Springer, **2019**. [doi]

[6] K. Roy, **A.R. Chowdhury**, A. Deyasi, and A. Sarkar. "Computing Surface Potential and Drain Current in Nanometric Double-Gate MOSFET Using Ortiz-Conde Model." In *Advances in Intelligent Systems and Computing*, Springer, vol. 812, pp. 41–47, **2019**. [doi]

[7] A. Deyasi, **A.R. Chowdhury**, K. Roy, and A. Sarkar. "Effect of High-K Dielectric on Drain Current of ID-DG MOSFET Using Ortiz-Conde Model." In *IEEE Electron Devices Kolkata Conference (EDKCON 2018)*, Kolkata, pp. 176–181, **2018**. [doi]

[8] **A.R. Chowdhury**, K. Roy, P. Saha, and A. Deyasi. "Analytical Investigation of Tunneling Current in Nano-MOSFET Using BSIM4 Model for Low Power VLSI Applications." In *Communications in Computer and Information Science (CCIS)*, Springer, vol. 836, pp. 11–18, **2018**. [doi]

Patent

[P1] S. Deb, R. Chowdhury, A. Sadhu, **A. Roy Chowdhury**, A.K. Dasgupta, A.K. Dhara, K. Roy, M. Bhattacharyya, and A. Chakrabarti. "Agonist-free system and method for detecting platelet function by stimulation of platelet aggregation." **Indian Patent IN580844 B1**, granted 17 February 2026. Application No. 201931048635, filed 27 November 2019. *Applicant: Sangutech Enterprises Private Limited.*

Awards & Achievements

- **UGC-NET (Lectureship)** qualified in Electronic Science — December 2020 & June 2021 (merged cycle).
- **Student Travel Grant** — CODS-COMAD 2024, IIT Jodhpur.
- **Student Travel Grant** — CODS-COMAD 2023, IIIT Bangalore.

Conference Presentations

- **A. Roy Chowdhury** and D.K. Sanyal. "Enhancing Indian Sign Language Translation via Motion-Aware Modeling." *WSLP @ AACL-IJCNLP 2025*, IIT Bombay, India. **2025**.
- A. Deyasi, **A.R. Chowdhury**, K. Roy, and A. Sarkar. "Effect of High-K Dielectric on Drain Current of ID-DG MOSFET." *IEEE EDKCON 2018*, Kolkata, India. **2018**.
- **A.R. Chowdhury** et al. "Tunneling Current in Nano-MOSFET." *52nd Annual Convention of CSI 2018*, Science City, Kolkata, India. **2018**.
- Student talk: "Fundamentals of Nano-Materials for Photonics Applications." Short-term course, NIT Durgapur, India. April **2018**.

Academic Service & Leadership

Administrative Chair — Winter School on Deep Learning (WSDL 2023) Jan – Mar 2023
ECSU, Indian Statistical Institute, Kolkata

- Coordinated a 9-week school with 15+ sessions on deep learning theory and applications; managed logistics, invited speakers, and hands-on lab sessions.
- Delivered hands-on session: *Convolutional Neural Networks with PyTorch* (virtual, 28 January 2023).

Organising Member — Winter School on Deep Learning (WSDL 2022) Jan – Mar 2022
ECSU, Indian Statistical Institute, Kolkata

– Delivered hands-on session: *Machine Learning — Regression & Clustering in Python*, Seventh Summer School on Computer Vision, Image Processing & ML, ISI Kolkata (3 & 7 June 2022).

Teaching

Teaching Assistantship. **COM 1101: Introduction to Computing** Autumn 2025
Indian Association for the Cultivation of Science (IACS), Kolkata

– Assisted with lectures, lab sessions, assignments, and assessment for the institute-wide introductory computing course.

Student Mentorship. Co-supervising the following Master's students with Dr. Debarshi Kumar Sanyal, SMCS, Indian Association for the Cultivation of Science, Kolkata:

– **Saumyadip Kar** — M.S. student, IACS Kolkata.
– **Shubhadeep Sarkar** — M.S. student, IACS Kolkata.

Technical Skills

Programming Languages	Python, MATLAB, C
Deep Learning	PyTorch, Scikit-learn
Computer Vision / NLP	OpenCV, Vision–Language Models, Sign Language Processing
Scientific Computing	NumPy, SciPy, Matplotlib
Tools & Platforms	Git, GitHub, Jupyter Notebook, LaTeX
Languages	English, Bengali, Hindi

References

Dr. Debarshi Kumar Sanyal
Associate Professor, SMCS, IACS Kolkata
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Prof. Swagatam Das
Professor, ECSU, ISI Kolkata
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Dr. Partha Basuchowdhuri
School Chair & Associate Professor, SMCS, IACS Kolkata
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Dr. Suryyani Deb
Asst. Professor (UGC), MAKAUT
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Dr. Ashis Kumar Dhara
Asst. Professor, Dept. of Electrical Engineering, NIT Durgapur
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